

Uniform Propagation Removes the Extra Diagonal Factor in Nonautonomous Maximal Regularity

Candidate partial result for arXiv:2403.00386

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Status: candidate partial result, likely valid. The general question in the source remains open. The extra polynomial factor is removed under an additional, natural propagation hypothesis, and the remaining exponential rate is shown to be optimal.

1 Source question

Let $I = (a, b)$, $T = b - a < \infty$, let $1 < p < \infty$, and let $D \hookrightarrow X$ be the common domain pair used in the source paper. For a nonautonomous maximally regular operator A , write

$$K = [A]_{\text{MR}_p(I)} = \|L_A^{-1}\|_{\mathcal{L}(L^p(I;X), \text{MR}_0^p(I))}, \quad L_A u = u' + A(\cdot)u.$$

Here $\text{MR}^p(I) = W^{1,p}(I; X) \cap L^p(I; D)$ has the source paper's time-homogeneous norm

$$\|u\|_{\text{MR}^p(I)}^p = \|u\|_{L^p(I;X)}^p + \|u\|_{L^p(I;D)}^p + T^p \|u'\|_{L^p(I;X)}^p,$$

and the subscript zero denotes $u(a) = 0$.

Belin and Lafitte prove

$$[A + \lambda]_{\text{MR}_p(I)} \leq c_p(\lambda T) [A]_{\text{MR}_p(I)},$$

where $c_p(\nu) \sim -\nu e^{-\nu}$ as $\nu \rightarrow -\infty$ and $c_p(\nu) \sim \nu$ as $\nu \rightarrow +\infty$. Immediately after Theorem 2.5 they ask whether the extra factor ν can be discarded; Remark 3.4 repeats that sharpness at either infinity is unknown [1].

(ii) $A \in \mathcal{MR}_p(I)$.

(iii) For any $\lambda \in \mathbb{R}$, $A + \lambda \in \mathcal{MR}_p(I)$.

And if one of these statements holds true, we have for any $\lambda \in \mathbb{R}$,

$$[A + \lambda]_{\mathcal{MR}_p(I)} \leq c_p(\lambda|I|)[A]_{\mathcal{MR}_p(I)} \quad (14)$$

where the function $c_p : \mathbb{R} \rightarrow \mathbb{R}_+$ satisfies

$$c_p(\nu) := (|\nu| + \kappa_p(\nu))(1 + e^{p\nu_-} - e^{-p\nu_+})^{\frac{1}{p}} \quad (15)$$

with $\frac{\kappa_p(\nu)}{|\nu|} \xrightarrow{|\nu| \rightarrow +\infty} 0$ and $\kappa_p(0) = 1$. In particular the following asymptotic behavior is observed for c_p :

$$\begin{aligned} c_p(\nu) &\underset{\nu \rightarrow -\infty}{\sim} -\nu e^{-\nu} \\ c_p(\nu) &\underset{\nu \rightarrow +\infty}{\sim} \nu \end{aligned}$$

It is not surprising to see the exponential factor $e^{-\nu}$ appear in the regime $\nu \rightarrow -\infty$ since this exponential behavior is already sharp for autonomous operators (see later Lemma 3.3). It is however still not clear to the authors whether one can discard the extra factor ν appearing. This asymptotic result is shown in Section 3. We can remark (14) possesses a particular invariance related to the behavior of $[A]$ with respect to the rescaling $A \mapsto \mu A(\mu \cdot)$ (see Remark 4.1 later). This is a direct consequence of structure of the time-homogeneous norm (5) for $\mathcal{MR}^p(I)$. In fact any estimate in the form

$$[A + \lambda]_{\mathcal{MR}_p(I)} \leq C(\lambda, |I|)[A]_{\mathcal{MR}_p(I)}$$

can be reduced to an estimate with $\tilde{C}(\lambda, |I|) := \inf_{\mu > 0} C(\mu\lambda, \frac{|I|}{\mu})$ thereby forcing the invariance of the inequality with respect to these transformations.

3.4.3. Maximal invariance of the sharpness estimate

Figure 1: Theorem 2.5 and the open sharpness remark in arXiv:2403.00386v2, PDF page 14.

2 Proof intuition

The source proof uses the scalar conjugation $u(t) = e^{-\lambda(t-a)}v(t)$ and estimates the full maximal-regularity norm of v before undoing the conjugation. Differentiating the scalar weight then produces the extra factor $|\lambda|T$.

If the unshifted problem has a uniformly X -bounded propagator, one can instead estimate only the X -norm of the shifted solution. Its Green kernel is the old kernel multiplied by $e^{-\lambda(t-s)}$, so Young's inequality gives the exact scalar L^1 factor. Returning to the unshifted equation,

$$L_A u = f - \lambda u,$$

then controls the entire maximal-regularity norm at once. This avoids differentiating the exponential weight inside the norm and hence avoids the extra polynomial factor.

3 Factor removal under uniformly bounded propagation on X

We isolate precisely the additional hypothesis used below.

Theorem 1 (Factor-free diagonal estimate). *Assume that $A \in \text{MR}_p(I)$ and that its zero-initial solution operator $R_0 = L_A^{-1}$ has a Green representation*

$$(R_0 f)(t) = \int_a^t U(t, s) f(s) ds \quad \text{for almost every } t \in I, \quad (1)$$

where $U(t, s) \in \mathcal{L}(X)$ is strongly measurable on $a < s < t < b$ and

$$M := \operatorname{ess\,sup}_{a < s < t < b} \|U(t, s)\|_{\mathcal{L}(X)} < \infty. \quad (2)$$

Then, for every real λ ,

$$\boxed{[A + \lambda]_{\text{MR}_p(I)} \leq (1 + M|1 - e^{-\lambda T}|)[A]_{\text{MR}_p(I)}}. \quad (3)$$

Consequently

$$[A + \lambda]_{\text{MR}_p(I)} = O(1) \quad (\lambda \rightarrow +\infty), \quad [A - \mu]_{\text{MR}_p(I)} = O(e^{\mu T}) \quad (\mu \rightarrow +\infty).$$

Thus the extra factor $|\lambda|T$ disappears in both regimes.

Proof. Fix $f \in L^p(I; X)$. The usual scalar conjugation, which is valid on the finite interval, shows that the unique solution of

$$u' + (A(t) + \lambda)u = f, \quad u(a) = 0, \quad (4)$$

is

$$u(t) = e^{-\lambda(t-a)} R_0(e^{\lambda(\cdot-a)} f)(t).$$

Using (1), this becomes

$$u(t) = \int_a^t e^{-\lambda(t-s)} U(t, s) f(s) ds. \quad (5)$$

Extend f by zero outside I and define

$$k_\lambda(r) = e^{-\lambda r} \mathbf{1}_{(0, T)}(r).$$

Equations (2)–(5), followed by the scalar Young convolution inequality, give

$$\|u\|_{L^p(I;X)} \leq M\|k_\lambda\|_{L^1(\mathbb{R})}\|f\|_{L^p(I;X)} = Mh_\lambda(T)\|f\|_{L^p(I;X)}, \quad (6)$$

where

$$h_\lambda(T) = \begin{cases} (1 - e^{-\lambda T})/\lambda, & \lambda \neq 0, \\ T, & \lambda = 0. \end{cases}$$

This quantity is positive for every real λ .

Now regard (4) as the unshifted equation $L_A u = f - \lambda u$. By the definition of $K = [A]_{\text{MR}_p(I)}$ and (6),

$$\begin{aligned} \|u\|_{\text{MR}_p(I)} &\leq K(\|f\|_{L^p(I;X)} + |\lambda|\|u\|_{L^p(I;X)}) \\ &\leq K(1 + M|\lambda|h_\lambda(T))\|f\|_{L^p(I;X)} \\ &= K(1 + M|1 - e^{-\lambda T}|)\|f\|_{L^p(I;X)}. \end{aligned}$$

The same formula at $\lambda = 0$ follows directly, and taking the optimal constant proves (3). Its two stated asymptotics are immediate. $\square$

Remark 2 (What the hypothesis means). *Hypothesis (1) is standard when A generates an evolution family on X which represents inhomogeneous zero-initial solutions, and (2) is uniform boundedness of that evolution family on the finite time triangle. The autonomous estimate in Lemma 3.3 of the source is the special case $U(t, s) = e^{-(t-s)A}$. Theorem 1 therefore extends that exact factor-free mechanism to a genuinely nonautonomous class.*

4 The exponential rate cannot be improved

The next elementary computation shows that the remaining negative-shift exponential in Theorem 1 has the correct order even in one dimension.

Proposition 3 (Scalar lower bound). *Let $I = (0, T)$, $X = D = \mathbb{R}$ with the same norm, and $A(t) = 0$. There is a constant $c_p > 0$, depending only on p , such that for every $\mu T \geq 2$,*

$$[-\mu]_{\text{MR}_p(0,T)} \geq c_p T e^{\mu T}. \quad (7)$$

In particular no estimate uniform in this class can replace $e^{\mu T}$ by a smaller exponential order as $\mu T \rightarrow \infty$.

Proof. Set

$$f_\mu(s) = \mu^{1/p} \mathbf{1}_{[0, 1/\mu]}(s).$$

Then $\|f_\mu\|_{L^p(0,T)} = 1$. The solution of $u' - \mu u = f_\mu$, $u(0) = 0$, is

$$u(t) = \int_0^t e^{\mu(t-s)} f_\mu(s) ds.$$

For $t \geq 1/\mu$ this gives

$$u(t) = (1 - e^{-1})\mu^{1/p-1}e^{\mu t}, \quad u'(t) = (1 - e^{-1})\mu^{1/p}e^{\mu t}.$$

Because $[T - 1/\mu, T] \subset [1/\mu, T]$ when $\mu T \geq 2$,

$$\begin{aligned} \|u'\|_{L^p(0,T)}^p &\geq (1 - e^{-1})^p \mu \int_{T-1/\mu}^T e^{p\mu t} dt \\ &= (1 - e^{-1})^p \frac{1 - e^{-p}}{p} e^{p\mu T}. \end{aligned}$$

The derivative term in the source norm therefore yields (7) with

$$c_p = (1 - e^{-1}) \left(\frac{1 - e^{-p}}{p} \right)^{1/p} > 0.$$

□

5 Scope and obstruction to the full question

Theorem 1 does *not* settle the source question for every $A \in \text{MR}_p(I)$. Bare maximal regularity gives the trace embedding

$$\text{MR}^p(I) \hookrightarrow C(\bar{I}; (D, X)_{1-1/p, p}),$$

so homogeneous propagation is naturally controlled on the trace space. It does not by itself supply a Green family uniformly bounded on the rough space X , which is exactly the input needed in (6). Attempting to use only the bounded Volterra inverse R_0 loses the pointwise time-triangle estimate; Neumann-series bounds then depend on the maximal-regularity constant in the exponent and do not recover the scale-sharp factor $e^{|\lambda|T}$.

Thus the remaining full problem can be stated sharply: determine whether the X -space convolution estimate behind (6), or an adequate substitute, follows from nonautonomous maximal regularity alone. Alternatively, a counterexample would need maximally regular coefficients whose propagation is controlled only on the trace space and for which the additional $|\lambda|T$ loss is genuinely attained.

The successful theorem was checked by two independent derivations: the scalar conjugation identity gives (5), while substitution into $L_A u = f - \lambda u$ gives the maximal-regularity estimate. Proposition 3 was checked by direct integration. A bounded search by arXiv id, exact open-question wording, and the phrases *nonautonomous maximal regularity*, *diagonal perturbation*, *evolution family*, and *sharp constant* located the source and later versions of the same paper, but no later resolution of the question or exact statement of Theorem 1. This is evidence, not a guarantee, of novelty.

References

- [1] T. Belin and P. Lafitte, *Quantitative estimates of L^p maximal regularity for nonautonomous operators and global existence for quasilinear equations*, arXiv:2403.00386v2 (2024); Electron. J. Differential Equations 2025, no. 18.